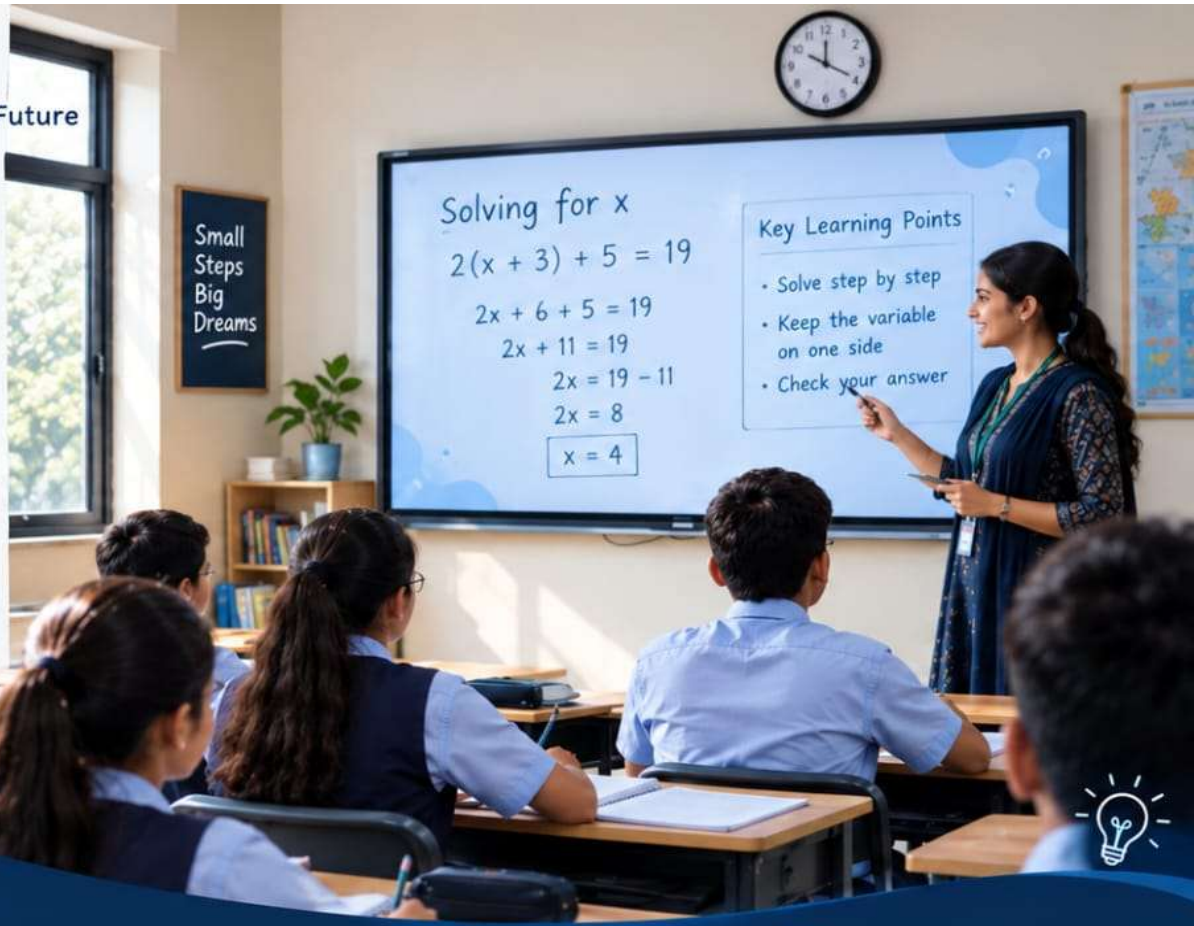
Personalized Learning | Better Progress | Brighter Future

Study Buddy

For Classes 6 – 10



Team Details

Team Leader
Amogh Mishra

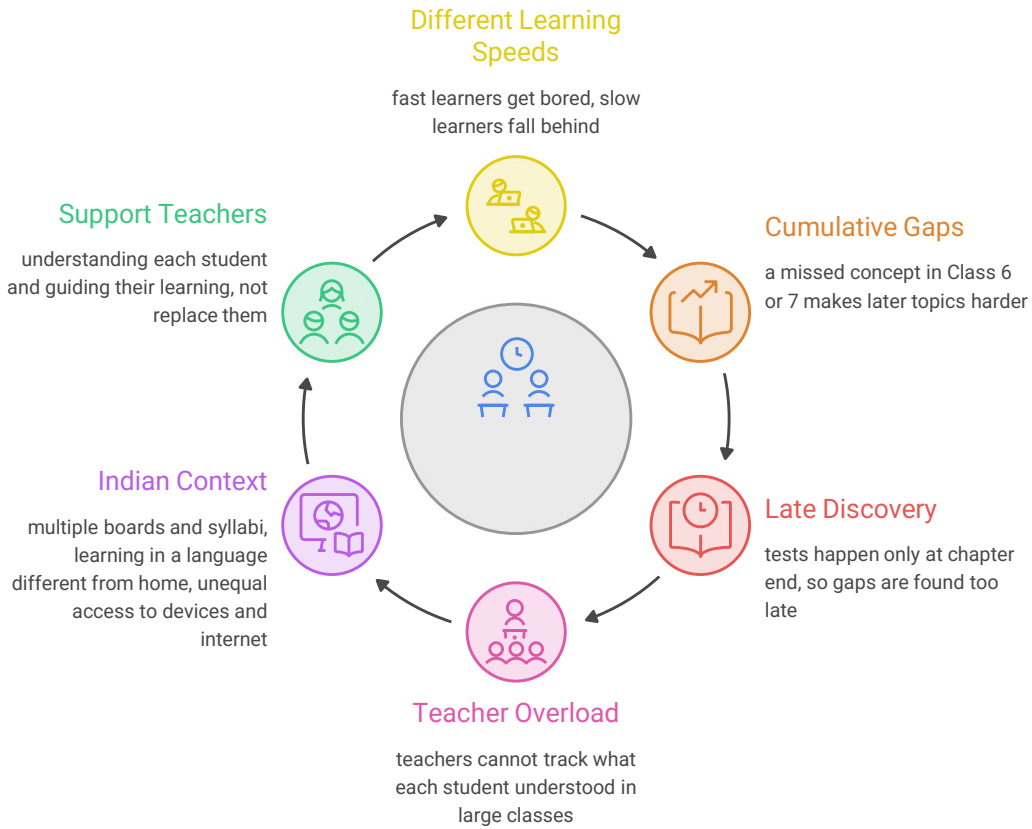
Phone No.
9335021508

Email Id
mishraamogh0807@gmail.com

No. of Team Members
3

College Name
Jaypee Institute of Information Technology, Noida

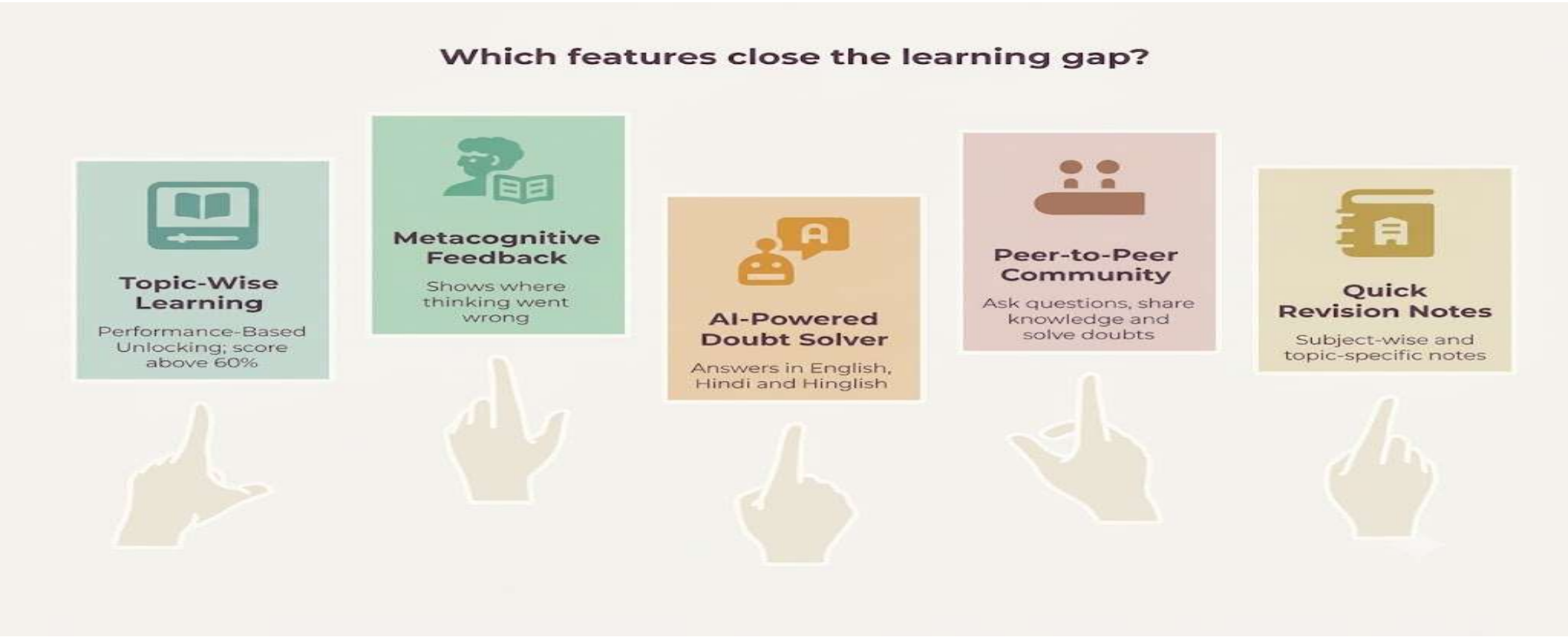
Adaptive Learning Problem Statement



Solution

Have you ever wondered why, in the same classroom, some students understand a concept instantly while others struggle with the very same topic? The problem is not that every student cannot learn; it is that **every student learns differently, yet education often follows the same pace for everyone.** To address this challenge, we are developing an **AI-powered adaptive learning platform** that personalises the learning journey for every student in Classes 6 to 10.

Our vision is simple: **not to make every student learn at the same pace, but to help every student learn at their own pace.**

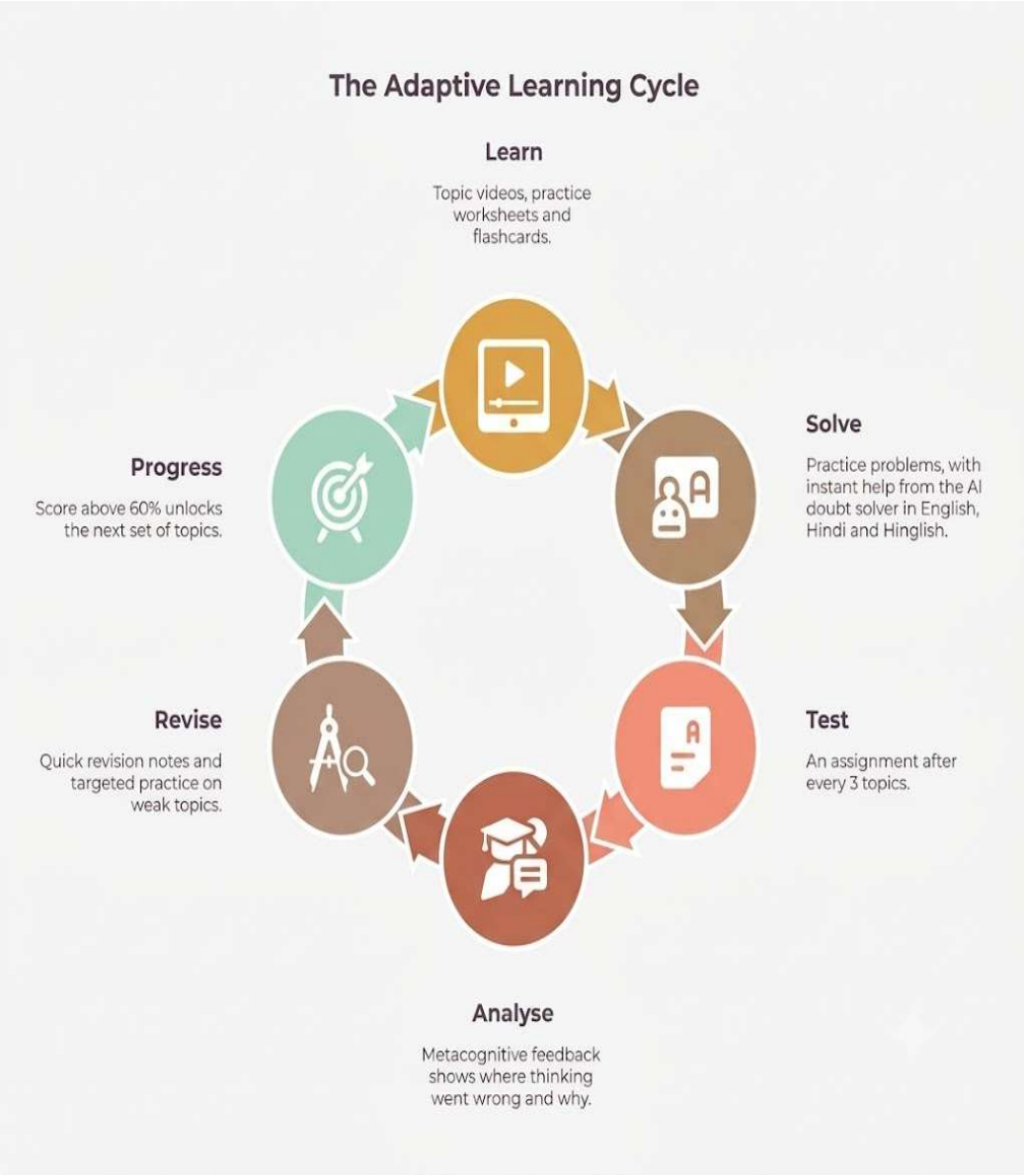


Which learning approach should be used?

Traditional Approach		Proposed Solution
The entire class moves at a single, fixed speed, regardless of individual differences. Uniform pacing	Pacing	Every student has a different path to success. Personalized learning path
Everyone gets the same lessons and questions, leaving fast learners bored and struggling students behind. Standardized content	Content and delivery	One concept is taught through different ways to understand. Multiple formats
Testing happens only at chapter end, so gaps are found long after the class moves on. Lagged assessments	Assessment timing	Master the concept before moving ahead. Mastery-based learning
Large classrooms offer little visibility into what each student understood. Overburdened teaching	Teaching support	Every doubt is solved and every concept understood. AI-powered doubt solver
Heavy reliance on coaching centres and generic apps adds extra hours without fixing foundational gaps. External patchwork	Learning support	Peer-to-gether and grow together. Peer-to-peer learning community

Innovation & Uniqueness

- **Earn-to-Advance Pathways:** Scores above 60% unlock the next topics; the rest get targeted revision first.
- **Beyond Marks, Into Mindset:** Feedback explains *where* and *why* a student's reasoning went wrong.
- **Hinglish-First Doubt Solving:** Instant help in English, Hindi, and Hinglish, the way students really speak.
- **One Loop, Not Five Tools:** Learn → Solve → Test → Analyse → Revise → Progress, fully connected.
- **Help That Fits the Need:** From quick notes to AI to peers, students get the lightest support that works.



✅ FEASIBILITY: WHAT WE BUILD FIRST

1 · 🧩 Lean, Modular Prototype

A small working core that proves the idea before it grows.

2 · 📱 Built for Shared Phones and Weak Internet

Lightest support first, so access never decides who learns.

🚀 SCOPE: HOW IT GROWS

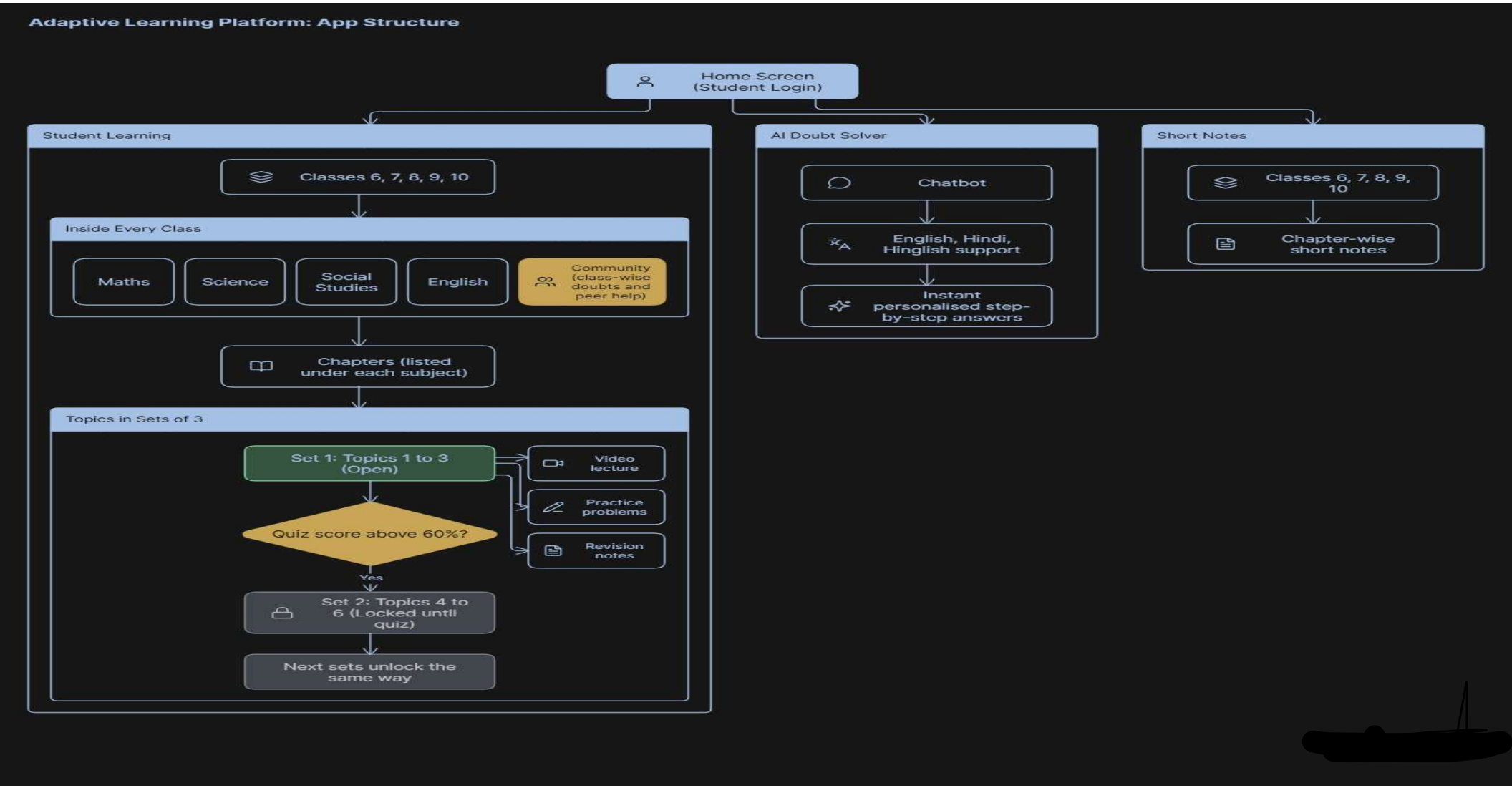
3 · 🌐 Curriculum and Language Expansion Path

Prove one chapter, then repeat the same template everywhere.

4 · 📊 Teacher Insight Dashboard

AI does the tracking so the teacher can do the teaching.

System Architecture



Our Technical Approach

Frontend Development
Utilize React and Streamlit for the user interface.

Database Management
Store data using Supabase and PostgreSQL.

Content Source
Reference NCERT and CBSE textbooks for educational material.

RAG-Based Doubt Solver
Answers only from NCERT text, in English, Hindi and Hinglish.

Low-Bandwidth Friendly
Web app (PWA) with text-first content for weak internet and basic devices.

Backend Infrastructure
Power the application using FastAPI and Python.

AI Integration
Implement the Gemini API for artificial intelligence.

Deployment Strategy
Deploy the application via Vercel and Render.

Mastery Engine
Score above 60% unlocks the next set of 3 topics.

